

## Mean



1 It measures the center/average of the scores

2

a 11.8

b 77.25

c 19.72

d -7.25

3 56

4 Greater

5 Smaller

6 19.87

7

a 1093.3

b 40

8

a 16

b 18

9

a

b 8.91

| Score ( $x$ ) | Frequency ( $f$ ) | $fx$ |
|---------------|-------------------|------|
| 5             | 14                | 70   |
| 7             | 4                 | 28   |
| 9             | 2                 | 18   |
| 11            | 18                | 198  |
| 13            | 6                 | 78   |
| Totals        | 44                | 392  |

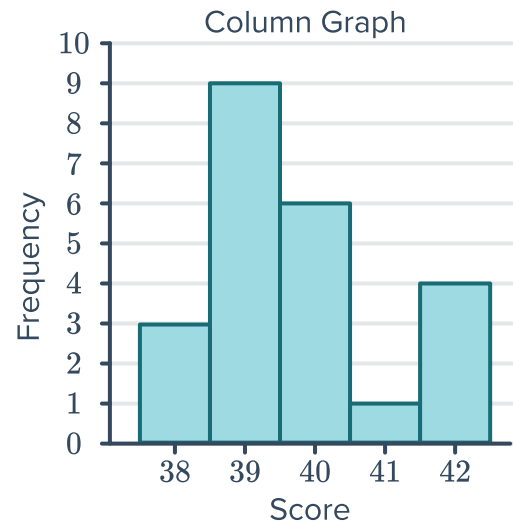
10

a

| Score ( $x$ ) | Frequency ( $f$ ) | $fx$ |
|---------------|-------------------|------|
| 38            | 3                 | 114  |
| 39            | 9                 | 351  |
| 40            | 6                 | 240  |
| 41            | 1                 | 41   |
| 42            | 4                 | 168  |
| Total         | 23                | 914  |



b



c 39.7

## Median

11 11

12

a 35

b Between 76 and 77

13  $n = 48$ 

14

a

i 46.6, 52.8, 63.9, 66.7, 69.4, 76.6, 80.6

ii 66.7

b

i  $-77.6, -54.8, -42.6, 40.6, 44.9, 45.6, 67.9, 74.7$ 

ii 42.75

c

i 1, 3, 11, 16, 16, 24, 24, 28

ii 16

15 37, 39, 41, 43

16

a 34

b 48

17



a

i 67

ii 45

iii 3062

iv 67

b

i 60

ii 4.5

iii 236

iv 3.9

18 83.5

19

a 25

b 18

c 13

20

a No

b No

c Yes

d No

## Mode

21

a 8, 14

b 26

c 3.0

d 71

22

a 74.07 hours

b 73 hours

c 67 hours

## Measures of center

23

a Mean, because there are no outliers.

b Mean, because there are no outliers.

c Mean, because there are no outliers.

d Median, because there is an outlier.

24

a Median = 657

b Mean = 108.8

25 \$1.64

26

a Mode = Median = 11

b Mean = Median = Mode = 90

c Median = 22

## Outliers



27 The mode will not change.

28 The outlier was smaller than the remaining values in the data set.

29

a \$527 000

b \$929 000

c \$446 600

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## Applications

30

a

| Heart Rate | Class center ( $x$ ) | Frequency ( $f$ ) | $f \times x$ |
|------------|----------------------|-------------------|--------------|
| 30 – 39    | 34.5                 | 13                | 448.5        |
| 40 – 49    | 44.5                 | 22                | 979          |
| 50 – 59    | 54.5                 | 24                | 1308         |
| 60 – 69    | 64.5                 | 36                | 2322         |

b 53.24

31

a 86

b 84.8

c 85.4

d 97, Student A

e 72, Student B

32

a \$399 400

b 40%

c \$386 500

d 50%

33  $x = 78$

34 63.65

35

a 15

b 13.29

c Median, because there is an outlier.

36

a 5 employees

b 58

c 20

d 36.5

e 20%



37

- a 90 minutes      b 25%      c \$80 000

38

a

| Weight in kilograms ( $x$ ) | Frequency ( $f$ ) | $fx$ |
|-----------------------------|-------------------|------|
| 16                          | 12                | 192  |
| 17                          | 17                | 289  |
| 18                          | 21                | 378  |
| 19                          | 10                | 190  |
| 20                          | 18                | 360  |
| 21                          | 22                | 462  |
| Total                       | 100               | 1871 |

- b 18.5 kilograms  
c 18.71 kilograms  
d Mean

39

- a 19      b 14 minutes      c 3 minutes      d 7.42 minutes  
e 5 minutes      f 7 minutes